

Zeke Naulty

Principal Software Engineer - AI Product & Platform (Synergy)

Principal Software Engineer — AI Systems, Full-Stack Architecture, & High-Reliability Platforms

Summary

Systems-first developer with a long arc in .NET and JavaScript, comfortable moving between desktop, web, and backend work to make long-lived products easier to change.

Focus areas include refactoring legacy code, tightening data flows, and shipping dependable features with clear UX and sensible engineering practices.

Experience

Principal AI Researcher & Engineer - EvoDev

Dec 2023 - Mar 2026 | Fairhope, AL

- Architected and implemented production-ready AI agent frameworks and RAG workflows using Python and C#/.NET, focusing on deterministic scope boundaries and traceable execution.
 - Developed the Domain of Domains (DoD), a formal runtime governance architecture for agentic systems addressing context drift and unverifiable memory in complex LLM environments.
 - Built BookForge, a Python-based AI workflow engine with a layered prompt governance pipeline, multi-provider LLM abstraction (OpenAI, Gemini, Ollama), and schema-validated state management.
 - Authored comprehensive technical decision records and architecture whitepapers for agentic coding and governance planes, ensuring knowledge sharing and system sustainability.
- Tech: Python, AI Agents, RAG (Retrieval-Augmented Generation), Vector Databases, System Architecture, Prompt Governance, C#/.NET, LLM Orchestration

Lead Developer / Technical Architect - iAdvantage Software, Inc.

Feb 2017 - Jan 2021 | Cary, NC

- Led the technical direction and delivery of a GLP-compliant lab platform in a highly regulated environment where data integrity and audit trails were critical requirements.
 - Shaped architecture and implementation across front-end (AngularJS/Bootstrap), APIs (REST/OData), and data (Oracle/PL-SQL) to support high-value product features.
 - Managed release engineering and Azure DevOps practices, ensuring reliability across both internet-connected and complex environment-specific deployments.
 - Mentored the development team through code reviews and technical guidance, maintaining high standards for production quality and maintainability.
- Tech: Full-stack Architecture, API Design, Relational Databases, Technical Leadership, Azure DevOps, Regulated Environments, TypeScript/Javascript

Senior Software Engineer (Backend & Services) - Storable

Jan 2021 - Dec 2022 | Cary, North Carolina, United States

- Architected .NET Core services using Entity Framework over PostgreSQL and SQL Server to drive high-availability insurance workflows.
 - Built background processing systems with Hangfire to synchronize eligibility and billing data across distributed web and desktop products.
 - Collaborated closely with product leadership to triage production issues and refactor high-risk architectural areas for long-term sustainability.
- Tech: C#, .NET Core, PostgreSQL, PostgreSQL Performance Tuning, Background Processing, Service Architecture

Principal Programmer / Analyst - Prestage Farms

Jan 2016 - Sep 2016 | Clinton, NC

- Led a team of engineers to modernize critical inventory and accounting systems, delivering high-value product features three months ahead of schedule.
- Designed and implemented role-based security and reporting engines for on-premise environments, ensuring operational stability during system cutovers.
- Balanced speed and quality by implementing pragmatic automation and bug-tracking pipelines that reduced

delivery bottlenecks.

Tech: Agile Leadership, System Modernization, Team Leadership, Security-Sensitive Environments, Requirement Shaping

Technical Architect - Solid Earth Inc.

Dec 2006 - Jan 2011 | Huntsville, AL

- Developed and optimized high-performance RETS client/server software on Oracle and C#, achieving industry-leading compliance and responsiveness.

- Architected configurable ASP.NET platforms where business rules and UI branding were driven by extensible data schemas, reducing individual code changes per customer.

Tech: Performance Tuning, High-Speed Data Processing, Oracle, Scalable Architecture

Skills

Core

- AI Agent Frameworks
- RAG (Retrieval-Augmented Generation)
- Python (AI/LLM)
- C# / .NET Core
- PostgreSQL
- System Architecture
- Full-Stack Engineering
- API Design
- Technical Leadership

Supporting

- Vector Databases
- TypeScript
- Docker & Linux
- On-Premise / Hybrid Deployment
- Azure DevOps
- Release Reliability
- Technical Documentation
- Regulated/Compliance Environments

Other

- Agile Methodologies
- AngularJS
- Bootstrap
- C#
- Cloud Platforms
- Cloud computing (Azure)
- CSS
- Database Development
- Full-stack development
- Git
- iOS
- JavaScript
- JIRA
- Kanban
- Leadership
- Mentoring
- Microsoft Azure
- Microsoft SQL Server
- Mobile application development
- Oracle
- Problem-solving
- Project management
- Regular Expressions

- Software development
- Team Building
- Team management
- Time management
- Unified Modeling Language (UML)
- Visual Basic .NET
- Web development
- .NET Core
- Entity Framework
- ASP.NET
- WinForms
- Hangfire
- Xamarin Forms
- OData/SOAP APIs
- Legacy modernization